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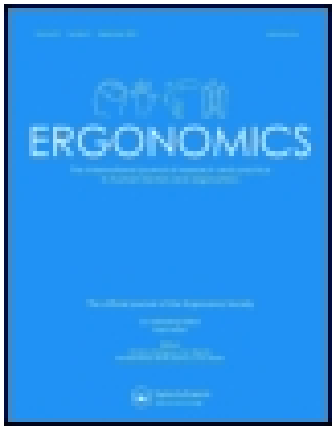

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Systems-based accident analysis in the led outdoor activity domain: application and evaluation of a risk management framework

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Systems-based accident analysis in the led outdoor activity domain: application and evaluation of a risk management framework

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Safety-compromising accidents occur regularly in the led outdoor activity domain. Formal accident analysis is an accepted means of understanding such events and improving safety. Despite this, there remains no universally accepted framework for collecting and analysing accident data in the led outdoor activity domain. This article presents an application of Rasmussen's risk management framework to the analysis of the Lyme Bay sea canoeing incident. This involved the development of an Accimap, the outputs of which were used to evaluate seven predictions made by the framework. The Accimap output was also compared to an analysis using an existing model from the led outdoor activity domain. In conclusion, the Accimap output was found to be more comprehensive and supported all seven of the risk management framework's predictions, suggesting that it shows promise as a theoretically underpinned approach for analysing, and learning from, accidents in the led outdoor activity domain.

Statement of Relevance: Accidents represent a significant problem within the led outdoor activity domain. This article presents an evaluation of a risk management framework that can be used to understand such accidents and to inform the development of accident countermeasures and mitigation strategies for the led outdoor activity domain.

Keywords: accident analysis; Accimap; human factors; outdoor activity; risk management

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Introduction

The outdoor education industry has expanded significantly over the past three decades. In the UK, for example, it is estimated that approximately three million school children are involved in adventure activities each year (Adventure Activities Licensing Authority 2009) and, in New Zealand, participation rates are reported to be at an all time high (Davidson 2004). Within the led outdoor activity domain (i.e. facilitated or instructed activities within outdoor education and recreation that have a learning goal associated with them), safety-compromising accidents and incidents represent a significant problem. Davidson (2004), for example, points out that serious accidents and incidents are occurring regularly and, in addition to the injury or loss of life and associated distress for those involved, such incidents generate negative publicity and have the potential to be extremely damaging to the domain (Davidson 2004). Despite this, compared to other safety-critical domains, relatively little is known about the causal factors involved in accidents and incidents. Further, little theoretically driven research has been undertaken to investigate what the causal factors are and how incident rates in the led outdoor activity domain can be minimised. This is despite the fact that such research has been undertaken in similar domains, such as the outdoor recreation and adventure tourism domains (e.g. Bentley *et al.* 2001, Bentley and Page 2008).

In other safety-critical domains, theoretically underpinned accident analysis is an accepted approach for identifying and understanding the causal factors involved in safety-compromising accidents and incidents and for informing the development of strategies and countermeasures designed to improve safety (Cassano-Piche *et al.* 2009). Despite recent model development (e.g. Davidson 2007), such approaches are yet to be developed and applied within a led outdoor activity context. This article presents an application of a systems-based risk management framework, widely used in other safety-critical domains, to the analysis of a well-known led outdoor activity accident. This involved the development of an Accimap (Rasmussen 1997) for the Lyme Bay sea canoeing incident, the aim being to test the usefulness of the approach for explaining how and why accidents occur in the outdoor activity domain. The outputs were then used to test seven predictions regarding safety and accident causation made by the framework. In addition, an existing framework, developed specifically for the led outdoor activity domain, was

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also used to analyse the accident and the output was compared to the Accimap output in order to compare both approaches.

Accidents in the led outdoor activity domain

Compared to other domains in which safety compromising accidents and incidents represent a significant problem, relatively little is currently known regarding such events within the led outdoor activity domain. A number of accident causation models have been developed specifically for this area (e.g. Meyer 1979 cited in Davidson 2004, Brackenreg 1999, Davidson 2007,); however, these models appear to be limited in terms of scope, theoretical underpinning and practical application. For example, in by far the most comprehensive work in the area to date, Davidson (2007) developed a root cause model of outdoor activity accident and incident causation and an associated error taxonomy for outdoor activity accidents and incidents (see Figure 1). According to the model, root causes emerging from poor safety management systems, or poor implementation of safety management systems, can in themselves lead to instructor errors that cause accidents and incidents or can lead to the instructors being placed in an unsafe environment or with unsafe students or with unsafe equipment (or combinations of the three), which, in turn, leads to accidents and incidents.

The model presented by Davidson (2007) represents the first real attempt to delineate, in detail, the causal factors involved in led outdoor activity incidents. In addition to the human-related root causes (overload, poor concentration, poor judgement, etc.), the model also acknowledges the role of safety management system failures in operator error and

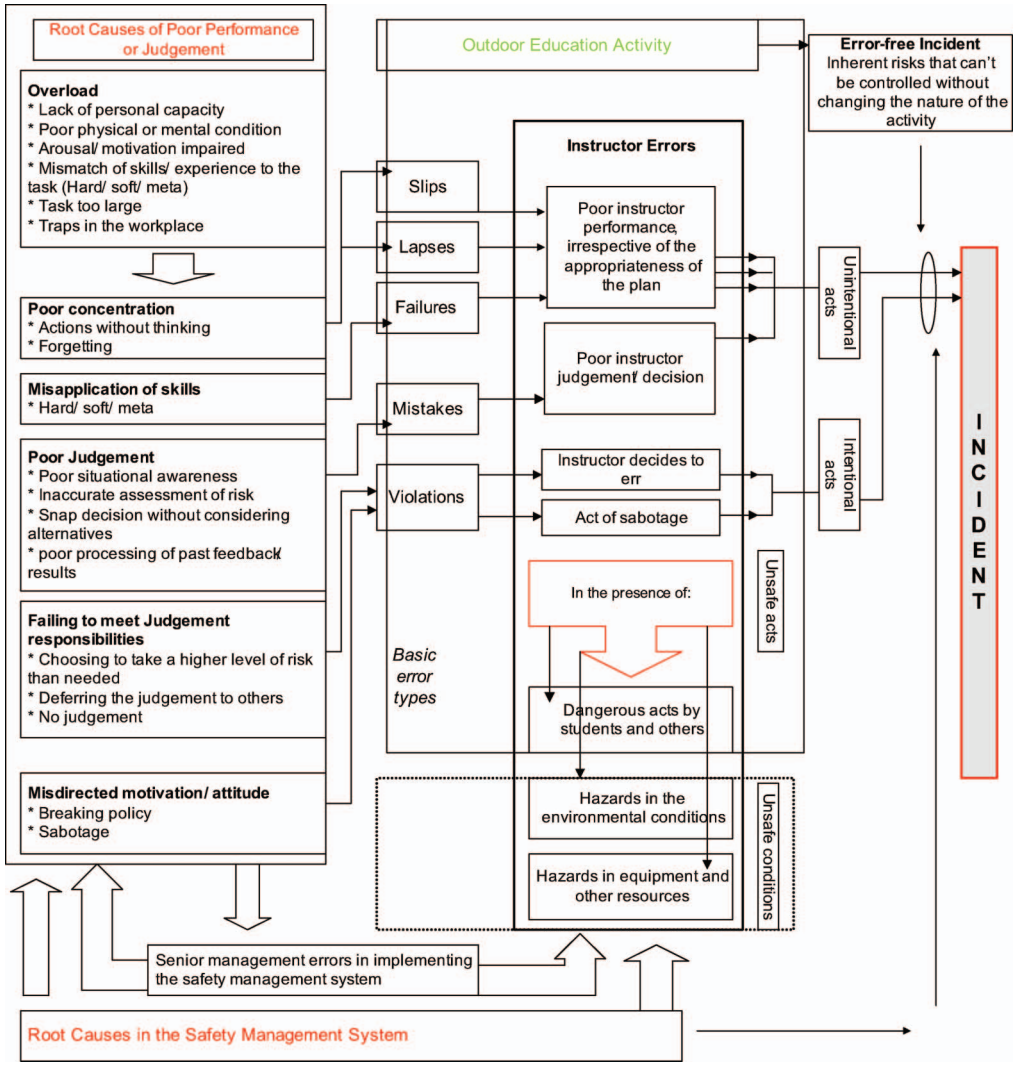

Figure 1. Root cause model of led outdoor activity accidents (adapted from Davidson 2007).

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accident and incident causation. The model has a number of significant flaws that raise questions over its application as an accident analysis tool in the led outdoor activity domain. For example, the model is based on only a limited set of accident and incident data (i.e. reports from only 20 instructors). As a corollary, the current root cause taxonomy appears limited. Taxonomies of different failures at the higher managerial system levels are also absent and no link is made between managerial failures and the root causes identified, which means that countermeasures cannot be developed for treating higher management failures. Finally, the model appears to lack a sound theoretical underpinning. For example, the term 'situation awareness' is cited as a root cause factor and yet much of the academic literature on situation awareness-related errors appears to have been ignored (e.g. Jones and Endsley 1996).

Systems-based accident causation models and analysis methods

Various models of accident causation exist (e.g. Reason 1990, Rasmussen 1997, Perrow 1999, O'Hare 2000, Leveson 2004). The most prominent of these are systems-based models (e.g. Reason 1990) and it is now widely accepted that the accidents that occur in complex socio-technical systems are caused by a range of interacting human and systemic factors (e.g. Reason 1990, Hollnagel 2004). Following the prominence of individual focused approaches to accident causation in the late 1970s, the systems approach to human error began to gain credence during the 1980s due to a series of catastrophes in which managerial failures were identified as key contributing factors, such as the Challenger, Piper Alpha, Hillsborough and Narita catastrophes (Johnson 1999). Following seminal theoretical work (e.g. Reason 1990), the systems approach is now the dominant approach to error and error management in most safety-critical domains. Accordingly, there exists a range of systems theory based accident analysis and investigation methodologies designed to allow investigators to identify the human and system-wide contributions to accidents and incidents occurring in complex socio-technical domains.

The most popular and widely applied model is Reason's (1990) systems perspective model of human error and accident causation. The 'Swiss cheese' model (presented in Figure 2), as it is more commonly known, focuses on the interaction between system-wide inadequate (referred to as latent) conditions and errors and their contribution to organisational accidents. Based on a layer by layer description of complex systems (e.g. decision makers, line management, productive activities and defences), the model purports that weaknesses in defences across all levels of the system, created by latent failures and unsafe acts, allow accident trajectories to breach defences and cause accidents.

The widespread acceptance of systems-based accident causation models, particularly Reason's, has led to the development of various systems-based accident analysis and investigation methods. Accident

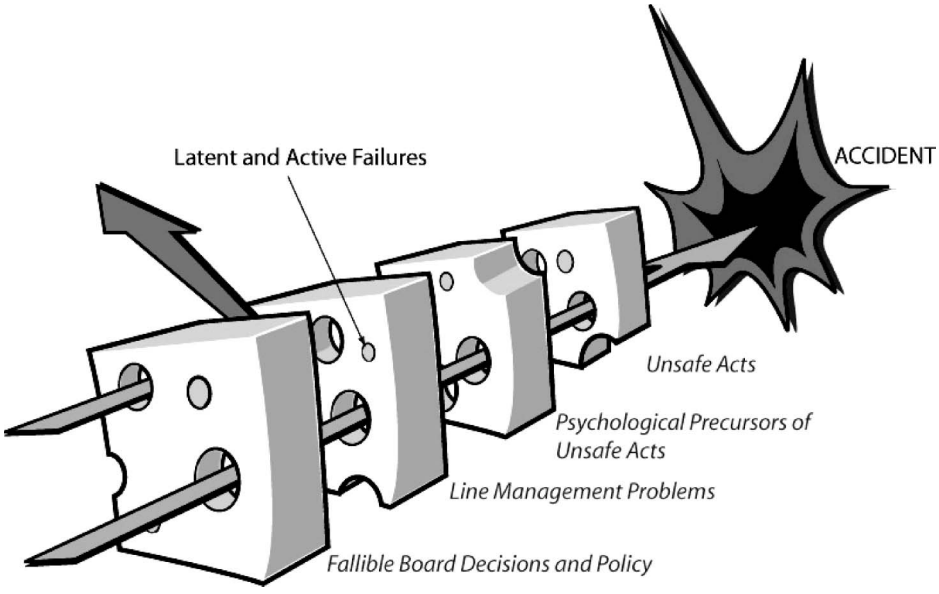

Figure 2. Reproduction of Reason's Swiss cheese accident causation model (adapted from Reason 2000).

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investigation is the most widely used method for clarifying the basic, contributing and immediate causes of accidents and for identifying appropriate measures to prevent occurrence of similar events in the future (Roed-Larson *et al*. 2004). Such methodologies are used to reconstruct accidents so that both the human and wider system contributions to accidents or incidents can be identified. The findings are then used to inform the development of measures to ensure similar accidents do not occur again. There are many systems-based accident analysis methods available; however, this article focuses on Rasmussen's risk management framework, which utilises the Accimap methodology.

Rasmussen's risk management framework

Also underpinned by systems-thinking, Rasmussen's risk management framework (Rasmussen 1997) is based on the notion that accidents are shaped by the activities of people who can either trigger accidental flows or divert normal work flows. Like Reason's model, Rasmussen's framework considers the various organisational levels involved in production and the control of safety and, in addition to errors and violations, focuses on the mechanisms generating behaviour within a dynamic work context. The model views complex socio-technical systems as comprising a hierarchy of actors, individuals and organisations (Cassano-Piche *et al*. 2009). Although the number of levels is not rigid and varies according to domain, the model typically focuses on the following levels: government; regulators; company; company management; staff, and work. Safety is viewed as an emergent property arising from the interactions between actors at each of the levels.

Like Reason, Rasmussen purports that various levels are involved in safety management via the control of hazardous processes through laws, rules and instructions. A representation of the levels involved in the control of safety within complex socio-technical systems is presented in Figure 3 (Rasmussen 1997). According to the framework, for systems to function safely, decisions made at high levels should promulgate down and be reflected in the decisions and actions occurring at lower levels of the system. Conversely, information at the lower levels (e.g. staff, work, equipment) regarding the system's status needs to transfer up the hierarchy to inform the decisions and actions occurring at the higher levels (Cassano-Piche *et al*. 2009). Without this so-called 'vertical integration', systems can lose control of the processes that they control (Cassano-Piche *et al*. 2009).

According to Rasmussen (1997), accidents are typically 'waiting for release', the stage being set by the routine work practices of various actors working within the system. Normal variation in behaviour then serves to release accidents. Thus, Rasmussen suggests that explaining accidents in terms of events, acts and errors has no utility for system redesign; rather, new approaches focusing on the mechanisms generating behaviour in the dynamic work context are required. For this purpose, Rasmussen (1997) proposes the Accimap approach.

The Accimap method

Rasmussen (1997) argues that task analytic methods are inadequate for analysing the accidents that occur in complex socio-technical systems; rather, a model of the behaviour shaping mechanisms involved is required. One approach proposed by Rasmussen is the Accimap modelling approach (Rasmussen 1997, Svendung and Rasmussen 2002), which is an accident analysis method that graphically represents the decision makers and decisions involved in producing the system in which an accident was permitted to occur. The Accimap method differs from typical accident analysis approaches in that it is used to identify and represent the causal flow of events upstream from the accident and also looks specifically at the planning, management and

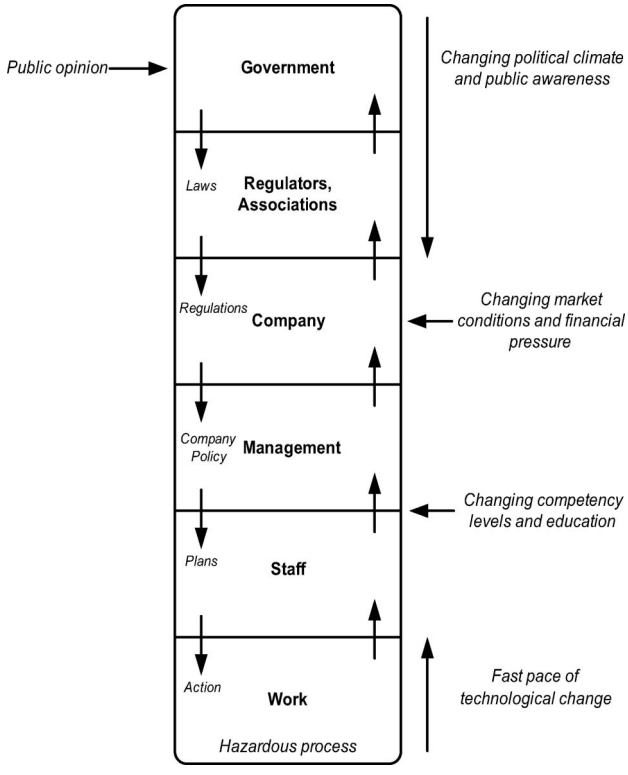

Figure 3. The levels of complex socio-technical systems involved in risk management (adapted from Rasmussen 1997).

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regulatory bodies that may have contributed to the accident (Svendung and Rasmussen 2002).

The method is underpinned by the notion that safety is an emergent property of complex socio-technical systems and is impacted by the decisions of all actors – politicians, chief executives, managers, safety officers and work planners – not just by those of front line operators alone (Cassano-Piche *et al.* 2009). The Accimap method therefore uses the following six organisational levels: government policy and budgeting; regulatory bodies and associations; local area government planning and budgeting (including company management); technical and operational management; physical processes and actor activities; equipment and surroundings). Failures at each of the levels are identified and linked between and across levels based on cause–effect relationships. Starting from the bottom of the graph, the equipment and surroundings level provides a description of the accident scene in terms of the configuration and physical characteristics of the landscape, buildings, equipment, tools and vehicles involved. The physical processes and actor activities level provides a description of the failures involved at the 'sharp end' of system operation (i.e. front line work activities). The remaining levels above the physical processes level represent all of the failures by decision makers that, in the course of the decision making involved in their normal work context, had or could have influenced the accident flow within the first two levels.

Due to its generic nature, the Accimap method is applicable in any domain and has previously been applied to a range of accidents and incidents occurring in various complex socio-technical systems, including gas plant explosions (Hopkins 2000), police firearm mishaps (Jenkins *et al.* 2010), loss of space vehicles (Johnson and de Almeida 2008), aviation accidents (Royal Australian Aviation Force 2001), public health incidents (Vicente and Christoffersen 2006, Cassano-Piche *et al.* 2009) and road and rail accidents (Svendung and Rasmussen 2002, Hopkins 2005).

Testing Rasmussen's risk management framework in the led outdoor activity domain

Rasmussen's framework makes a series of predictions regarding accidents in complex socio-technical systems. These predictions have previously been used to evaluate the framework in a number of domains, including food production (Cassano-Piche *et al.* 2009), water distribution (Vicente and Christoffersen 2006) and armed police response (Jenkins *et al.* 2010). Similarly, in addition to evaluating the utility of the Accimap method for the analysis of led outdoor accidents and incidents, the predictions made were used to test Rasmussen's framework in the context of accidents and incidents in the led outdoor activity domain. The predictions are presented in Table 1.

Case study analysis: the Lyme Bay Incident

The Lyme Bay canoeing incident was used for this case study. The primary justification for this is the quality and depth of the data available for the Lyme Bay incident (i.e. the inquiry report), which supported the in-depth analysis required for accident model comparison. Initially, the study was to involve a subset of incidents; however, a preliminary analysis of a series of cases (provided by the project stakeholders) indicated that there was insufficient data available to support the level of analysis required for accident model comparison.

Incident description

The Lyme Bay sea canoeing incident involved the death of four students whilst on an outdoor education activity trip on 22 March 1993. The activity involved a

Table 1. Rasmussen's risk management framework's predictions regarding accidents in complex sociotechnical systems*.

Prediction
1. Safety is an emergent property of a complex socio-technical system. It is impacted by the decisions of all of the actors—politicians, managers, safety officers and work planners— not just the front-line workers alone.
2. Threats to safety or accidents are usually caused by multiple contributing factors, not just a single catastrophic decision or action.
3. Threats to safety or accidents can result from a lack of vertical integration (i.e. mismatches) across levels of a complex socio-technical system, not just from deficiencies at any one level alone.
4. The lack of vertical integration is caused, in part, by a lack of feedback across levels of a complex socio-technical system. Actors at each level cannot see how their decisions interact with those made by actors at other levels, so the threats to safety are far from obvious before an accident.
5. Work practices in a complex socio-technical system are not static. They will migrate over time under the influence of a cost gradient driven by financial pressures in an aggressive competitive environment and under the influence of an effort gradient driven by the psychological pressure to follow the path of least resistance.
6. The migration of work practices can occur at multiple levels of a complex socio-technical system, not just one level alone.
7. Migration of work practices causes the system's defences to degrade and erode gradually over time, not all at once. Accidents are released by a combination of this systematically- induced migration in work practices and a triggering event, not just by an unusual action or an entirely new, one-time threat to safety.

*Adapted from Cassano-Piche *et al.* 2009.

group of eight students, a school teacher, a junior instructor and a senior instructor engaging in an introductory open sea canoeing activity in Lyme Bay, Dorset, UK. After a series of initial capsizes, the junior instructor and eight students became separated from the senior instructor and their school teacher and were blown out to sea. Due to high wind and wave conditions, each canoe in the junior instructor/student group was subsequently swamped and capsized. The junior instructor and students were left in the water, with all canoes abandoned. Four students drowned as a result. The ensuing inquiry identified a number of failures on behalf of the company providing the activity and also the instructors overseeing the activity, including that the activity was not suitable for novices, that the instructors were not adequately qualified to oversee the activity and that various pieces of key equipment were not provided, including spray decks, flares, towlines and a survival bag.

Methodology

Both the Accimap methodology and the root cause model were used to model the Lyme Bay incident. Two human factors researchers performed the analyses independently of one another, each using one of the methods. The analyses were based on the official inquiry report into the incident (Jenkins and Jenkinson 1993), which provided an exhaustive account of the incident and the failures involved. The process involved each analyst reviewing the official inquiry report and then using their allocated method (e.g. Accimap or root cause method) to classify the different failures that led to the accident occurring. A continual process of reviewing and refining the analyses was used by each analyst. In order to ensure reliability and validity, three additional human factors researchers independently reviewed the analyses and input data. Any subsequent disagreements were resolved through further discussion and consensus between the analyst and researcher in question. For the validation process, the analyses and input data were sent by the lead author, via email, to the researchers in question. Following this, an expert panel of outdoor education and adventure activity providers independently reviewed and refined the analysis outputs (in consultation with all human factors researchers involved). Again, the analysis outputs and input data were sent by the lead author, via email, to the panel members.

Results

The Accimap for the Lyme Bay incident is presented in Figure 4. A representation of the root cause model classification is presented in Figure 5.

Accimap results

Equipment and surroundings

The inquiry report cited the failure of the activity centre and instructors involved to provide appropriate equipment for the canoeing activity as a key contributing factor. There were three prominent failures at this level, including a failure to provide equipment, the provision of inappropriate equipment or the incorrect use of equipment. The inquiry report identifies the fact that spray decks, the device that creates a watertight seal at the point where the user sits in the canoe, were not provided in the students' canoes, despite the students all being inexperienced in the operation of canoes and also in sea canoeing in general (the instructors used spray decks). This was cited as a key causal factor in the ease in which the canoes were swamped and capsized. This is not, however, as surprising as it first appears; often at the introductory level, spray decks can be a learning impediment. More in question would be the decision to take beginners into an area where spray decks were necessary. Further, unlike the instructors, the students' canoes did not have supplementary buoyancy (such as sealed bags filled with air, which is generally recommended for sea canoeing) and were not fitted with deck lines, which meant that they could not be tied to each other or towed in the event of the group becoming separated. All students were inadequately clothed for sea canoeing and the lifejackets worn by the students were not inflated, on instruction from the instructors, and had no whistles attached to them. According to the inquiry report, inflation of lifejacket is the recommended step for preventing drowning, since it raises the wearer's head out of the water.

Other equipment that would normally be taken on sea canoeing trips, but was not taken in this case, included radios, distress flares, towlines and a survival bag. The absence of towlines meant that the group was not connected and allowed the students to drift out to sea. However, the lack of towlines is probably down to their limited use (towing students with towlines in practice is often very difficult and can be unsuccessful). Without radios and distress flares, the alarm could not (quickly) be raised by instructors or students. Together with the instructors, the school teacher on the trip, also an inexperienced canoeist, was given a Lazer three fifty canoe (Jenkins and Jenkinson 1993), which was shorter and lighter than the other canoes and deemed inappropriate for use by a novice canoeist at sea. Finally, at this level the conditions also played a role; whilst the wind speed level was not overly high, the offshore winds and high seas were key factors in sweeping the students out to sea.

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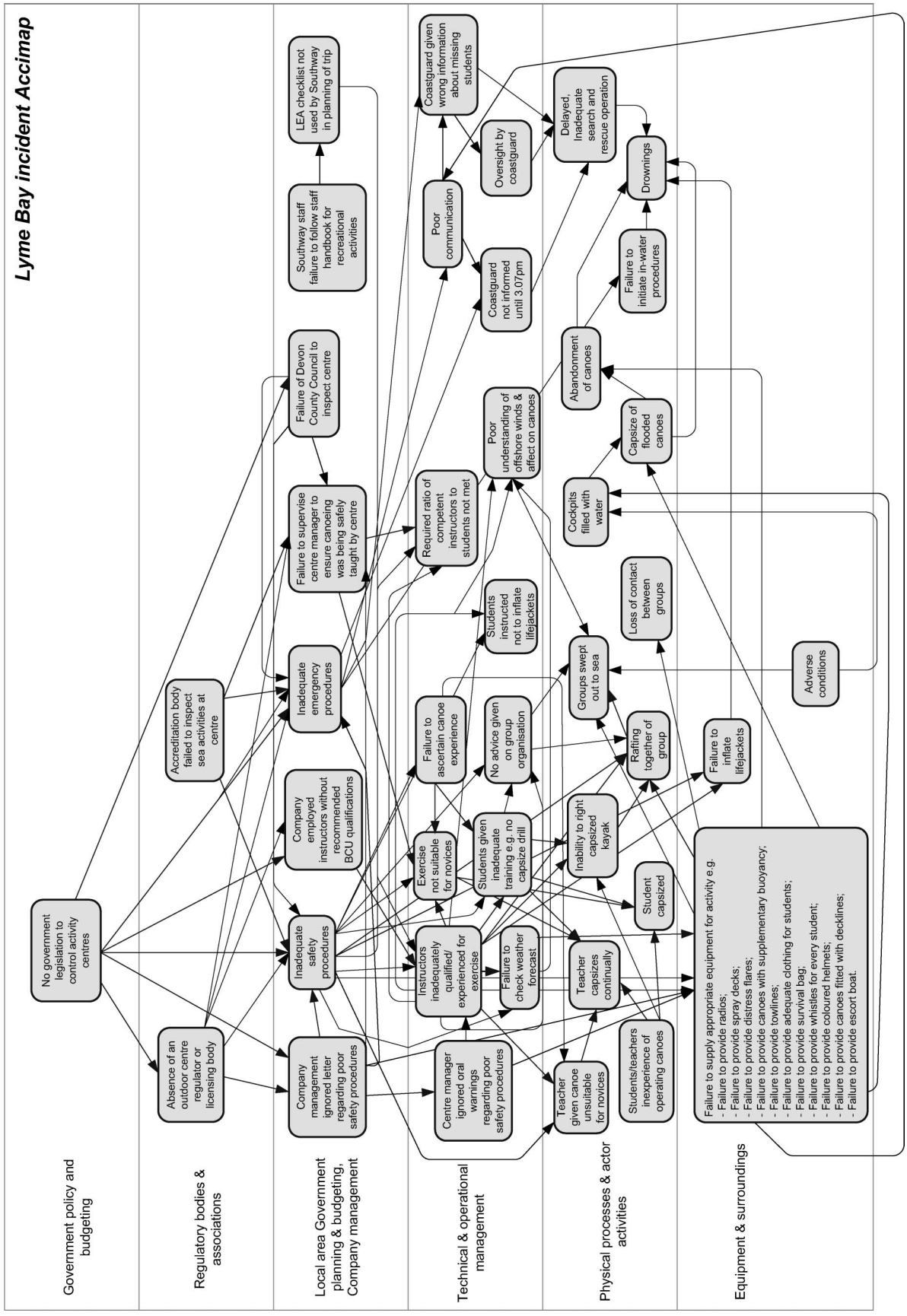

Figure 4. Accimap, showing the contributory factors involved in the Lyme Bay canoeing incident.

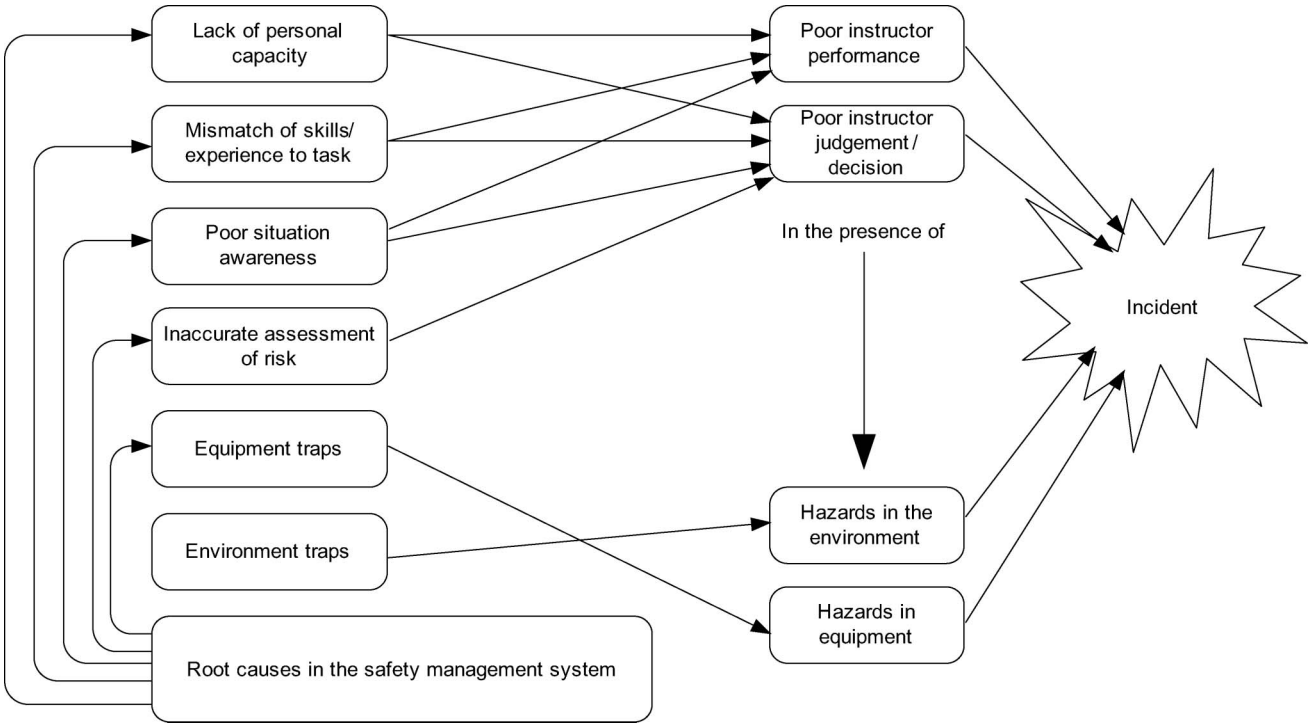

Figure 5. Classification of failures involved in the Lyme Bay canoeing incident using Davidson's (2007) root cause model.

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Physical processes and actor activities

At the physical processes and actor activities level, most of the contributing factors relate to the instructors' and students' inability to respond to the unfolding situation, which was a function of their lack of experience and qualifications for such an activity and the absence of appropriate safety and emergency procedures. First, the initial, continued capsizes of the teacher's canoe and the collective inability of the senior instructor and the teacher to successfully right the capsized canoe were the trigger events. Whilst the senior instructor attempted to right the capsized canoe, the junior instructor proceeded to raft together the students (i.e. organise tightly together in formation), although prior to this no advice had been given to the students on group organisation. Influenced by the offshore winds, the rafted group was quickly swept out to sea and, due to the absence of distress flares and radios, the two groups (instructor/teacher group and rafted group, including eight students and one junior instructor) lost contact with each other. Due to the absence of tow or deck lines, the rafted group could not be tied together. As the students were swept further out to sea, the increasingly adverse conditions (e.g. higher waves), coupled with the lack of spray decks, led to further capsizes and the canoes becoming swamped with water until eventually all of the students were submerged in the water with only one upturned canoe on to which they could hold. Eventually, after a failed attempt to paddle ashore using the upturned canoe, the final canoe was abandoned. Whilst in the water, standard in-water procedures, designed to keep submerged entities warm and prevent hyperthermia, were not initiated.

The activities undertaken in organising a response were also important. The alarm was not raised onshore until the group was over 3 h late and, initially, the site manager did not immediately notify the coastguard; rather, he spent some time himself searching the shoreline for the missing canoeists. In addition, the coastguard was wrongly informed that the instructors were well qualified, over 18 years of age and were well equipped for the activity, all of which was not the case. Once the rescue was underway, the coastguard themselves made a number of errors, although this is not the focus of the present analysis.

Technical and operational management

At the technical and operational management level, decisions and actions both preceding the accident and on the day of the activity played a significant role in the failures at the lower levels. Prior to the incident, the centre's manager (and also the company management) failed to heed the content of a letter sent by two previous employees to management regarding unsafe practices, poor safety procedures and inadequate

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equipment for the sea canoeing activities provided at the centre. The centre manager also failed to heed verbal warnings from the same two employees regarding the fact that canoeing was not being safely taught at the centre (Jenkins and Jenkinson 1993). The instructors employed by the centre were also not sufficiently qualified for sea canoeing, which represents a failure on behalf of the manager to employ adequately qualified staff. The exercise itself was not suitable for novice canoeists; further, the inquiry report suggests that no attempt was made by staff to ascertain the experience levels of the students or teacher. As a result, the training given to the teacher and students prior to the trip in the centre's swimming pool was inadequate, involving no capsize drills whereby the procedure to right and re-board canoes are taught. This is particularly important with regard to the inability of the teacher to right the capsized canoe, which was one of the key events that led to the group becoming separated.

On the day of the activity, neither of the two instructors checked the weather forecast and no attempt was made to ascertain the conditions out at sea. Students were also told not to inflate their lifejackets even though inflation of lifejackets is a standard approach to avoid drowning. Due to the instructors' lack of qualifications and the type of activity being undertaken, the required ratio of competent instructors to students was not met.

Local area government planning and budgeting, company management

At the company management level, the company's management also failed to heed the content of the letter sent by two previous employees regarding the safety of canoeing activities at the centre in question. The inquiry also reported that the managing director was charged with failing to devise and enforce safe procedures for executing sea canoeing activities and that the emergency procedures in place were also inadequate. The employment of inadequately qualified instructors is also represented at this level, with the company management failing to procure employment of staff suitably qualified to provide safe canoeing activities. There was also a failure on behalf of company management to supervise and ensure that safe activities were being provided at the centre in question. Finally, in what was a controversial finding at the time, the inquiry report also concluded that the school involved had not planned the trip adequately. According to the inquiry report, the school in question failed to adequately follow their own staff handbook for organising such trips and a local education authority checklist for planning such activities was not used.

Regulatory bodies and associations

At the time of the incident, there was no regulatory body or licensing body for outdoor activity centres. This meant that unsafe practices and procedures in the led outdoor activity domain could, to a large extent, continue unchecked without consequence. Following the incident, the inquiry report recommended that a national independent system of registration and regulation of outdoor activity centres be developed as a matter of urgency and also that no school or youth group should be permitted to use such centres until they had been approved for registration by an appropriate body. The absence of a regulator and legislation undoubtedly allowed the company management to continue with unsafe and inappropriate procedures and to employ inadequately qualified staff. The report concluded that, if a regulatory body had been in place at the time, the concerns reported by previous employees would undoubtedly have been reported to the body and appropriate action would have ensued.

In addition, although the centre in question was accredited by the British Activity Holidays Association, the inquiry report suggested that no examination of sea canoeing activities was undertaken; rather, only land- and pool-based activities were examined.

Government policy and budgeting

At the government policy and budgeting level, the absence of legislation to control outdoor activity centres is the key failure involved. This meant that there was no regulating or licensing body overseeing outdoor activity providers, which enabled the centre to continue engaging in unsafe practices, despite these being identified and documented by previous employees of the centre.

Root cause model results

The root cause model taxonomy identified a number of root causes of instructor error. Equipment traps were present and were created by the failure of the activity centre and instructors involved to provide appropriate equipment for the canoeing activity. The equipment traps included a failure to provide equipment, the provision of inappropriate equipment or the incorrect use of equipment (see Accimap results section for description of equipment failures). The root cause model also highlighted the role of environment traps, which, in this case, were the adverse conditions (e.g. high wind and waves) out at sea and also the choppy sea conditions, which played a role in the initial capsizes.

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The root cause model identified a range of root causes for inadequate instructor performance and/or judgement and decision making. The first root cause identified was a mismatch of skills and experience to the task. The instructors were found to be inexperienced and inadequately qualified for the open sea canoeing activity. This root cause also applies to the students, who were inexperienced in open sea canoeing. Second, inaccurate assessment of risk was identified on behalf of the instructors, since they did not appreciate the risks associated with taking inexperienced students out on an open sea canoeing activity. Lack of personal capacity for the activity was also identified as a root cause. The task was beyond the personal capacity of the inexperienced instructors, particularly when the situation began to go wrong (i.e. capsizes). Finally, poor situation awareness on behalf of the instructors was also identified as a root cause.

At the managerial level, the model identified root causes in the managerial systems employed by the company involved. These included inadequate procedures and inadequate emergency procedures. The inquiry reported that the managing director was charged with failing to devise and enforce safe procedures for executing sea canoeing activities and that the emergency procedures in place were also inadequate. The instructors employed by the centre were also not sufficiently qualified for sea canoeing or canoeing, which represents a failure on behalf of the managerial system to employ adequately qualified staff.

Discussion

The aim of the analysis presented was, first, to test Rasmussen's risk management framework as a method for analysing accidents and incidents in the led outdoor activity domain and, second, to compare the outputs of the risk management framework analysis with an analysis derived from a framework developed specifically for the led outdoor activity domain. The Lyme Bay tragedy, a high profile incident in the area, was used as a case study incident for this purpose.

In order to test the utility of Rasmussen's framework for explaining such incidents, the Accimap approach was used to ascertain whether the contributory events involved supported each of the predictions made by Rasmussen's framework. This process has been used previously to test the framework in other safety-critical domains, such as food production (Cassano-Piche *et al*. 2009), water distribution (Vicente and Christoffersen 2006) and armed police response operations (Jenkins *et al*. 2010). The predictions outlined for testing the risk management framework were all supported by the case study analysis. Table 2 gives an overview of the predictions and examples of how the analysis supported them.

The results of this case study therefore provide further support for the Accimap approach as a valid and appropriate accident analysis methodology for use in the led outdoor activity domain. These findings add to the considerable evidence presented in the literature regarding the utility of Rasmussen's framework (e.g. Vicente and Christoffersen 2006, Cassano-Piche *et al*. 2009, Jenkins *et al*. 2010).

The analysis presented further demonstrates the utility of the Accimap approach for analysing, and learning from, accidents and incidents in the complex, safety-critical domains. The approach is generic and can be applied in any domain (the led outdoor activity domain being an example of a domain in which the application of systems-based accident analysis methods has previously not received significant attention), is simple to apply and the output is easily interpretable. The identification of factors across the range of systemic levels allows the development of appropriate systems-focused countermeasures and remedial strategies, as opposed to individual oriented measures that ignore the wider systemic causal factors often produced from individual 'operator at the sharp end' accident analyses (Reason 1997, Dekker 2002). That said, the approach also caters for the unsafe acts made by operators at the 'sharp end'. Using the approach to model accidents and incidents in the outdoor activity domain is therefore likely to yield useful insights that identify ways in which the likelihood of future safety-compromising accidents and incidents can be reduced.

The Accimap method does have some limitations. While comprehensive, the approach is time-consuming in application and requires significant resources for data collection and analysis activities. In addition, classification schemes providing different error and failure types at each of the different levels are not available and so different analysts often use different terminology and/or classify failures differently. The absence of classification schemes at this level may also lead to analysts missing failures or errors, since they are given no assistance by the methodology in the process of identifying them. The approach could also be criticised for a lack of focus on the physical environment and context; however, with appropriate analysts this should be avoided. As with all accident analysis approaches, the quality and validity of the analysis produced is ultimately dependent upon the quality of the data available and of the analyst(s) undertaking the analysis. For example, for this approach to become an industry standard in the led outdoor activity domain, development of appropriate analyst training programmes and data collection and

Table 2. The results of the testing of the risk management framework's predictions.

Prediction and case study evidence
1. Safety is an emergent property of a complex socio-technical system. It is impacted by the decisions of all of the actors—politicians, managers, safety officers and work planners— not just the front-line workers alone. The Accimap presented in Figure 4 demonstrates that actors at all levels of the system were involved in creating the conditions that led to the drowning incident. This is represented by the presence of failures at each of the levels specified by Rasmussen's framework.
2. Threats to safety or accidents are usually caused by multiple contributing factors, not just a single catastrophic decision or action. The Accimap presented in Figure 4 demonstrates that no one factor identified was independently responsible, in isolation, for the drowning incident. For example, failure of the students' canoes to be fitted with spray decks can be attributed to a range of factors, including the company's employment of inexperienced, inadequately qualified instructors, the centre and company managements failure to heed verbal and written warnings from previous employees of the centre and the absence of appropriate safety procedures at the centre in question.
3. Threats to safety or accidents can result from a lack of vertical integration (i.e. mismatches) across levels of a complex socio-technical system, not just from deficiencies at any one level alone. The Accimap presented in Figure 4 demonstrates that there were various mismatches between individuals and organisations across the different levels of the system responsible for delivering safe and efficient outdoor activities.
4. The lack of vertical integration is caused, in part, by a lack of feedback across levels of a complex socio-technical system. Actors at each level cannot see how their decisions interact with those made by actors at other levels, so the threats to safety are far from obvious before an accident. The Accimap presented in Figure 4 demonstrates how decisions and actions made at the different levels were made without appreciation of, or feedback regarding, how they interacted with those made by actors at other levels. For example, the provision of sea canoeing activities at the centre was made without any acknowledgement or consideration of the other levels of the system. Feedback levels regarding the performance of such activities were not forthcoming from the higher levels, including company management and the accrediting body (who did not examine sea-based activities).
5. Work practices in a complex socio-technical system are not static. They will migrate over time under the influence of a cost gradient driven by financial pressures in an aggressive competitive environment and under the influence of an effort gradient driven by the psychological pressure to follow the path of least resistance. Economic and efficiency pressures played a part in the decisions made and actions taken by those involved, the result of which was the system migrating across the safe boundary of operation. Economic pressures were extant predominantly at the local area government and company management level and efficiency pressures mainly affected the company management and technical and operational management levels.
6. The migration of work practices can occur at multiple levels of a complex socio-technical system, not just one level alone. In this case the migration of work practices occurred at four different levels: the local area government and company management level; technical and operational management level; the physical processes and actor activities level; the equipment and surroundings level.
7. Migration of work practices causes the system's defences to degrade and erode gradually over time, not all at once. Accidents are released by a combination of this systematically- induced migration in work practices and a triggering event, not just by an unusual action or an entirely new, one-time threat to safety. Some of the practices that contributed to the accident had clearly been affecting the system for a significant portion of time prior to the accident occurring. The letter from previous employees regarding poor safety procedures and equipment in the context of sea canoeing activities, sent 6 months prior to the accident, is clear evidence that migration of work practices had been occurring for some time.

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storage systems would be required. In addition, significant resources would be required for accident and incident analyses and it is notable that the method provides no guidance in the development of remedial measures and countermeasures. Also, as with most accident analysis approaches, the Accimap method suffers from various problems associated with hindsight. Dekker (2002), for example, suggests that hindsight can potentially lead to oversimplified causality and counterfactual reasoning.

The second aim of this article was to compare the Accimap output with that provided by a root cause model developed specifically for the led outdoor activity domain. For this purpose, the Accimap output was compared with that derived from the analysis supported by Davidson's (2007) root cause model (Figure 5). The analyses presented highlight the viewpoint on accident causation provided by both approaches. The Accimap method provides an analysis of the failures across the overall system, comprising six levels, ranging from government and regulator levels to front line operator and equipment levels. The root cause model, on the other hand, covers only the front line operators, with its root causes taxonomy and management levels, albeit only with reference to managerial system failures and without identifying the linkages between failures across different levels. This is represented in Figure 6, which shows a representation of each analysis mapped on to Rasmussen's risk management framework.

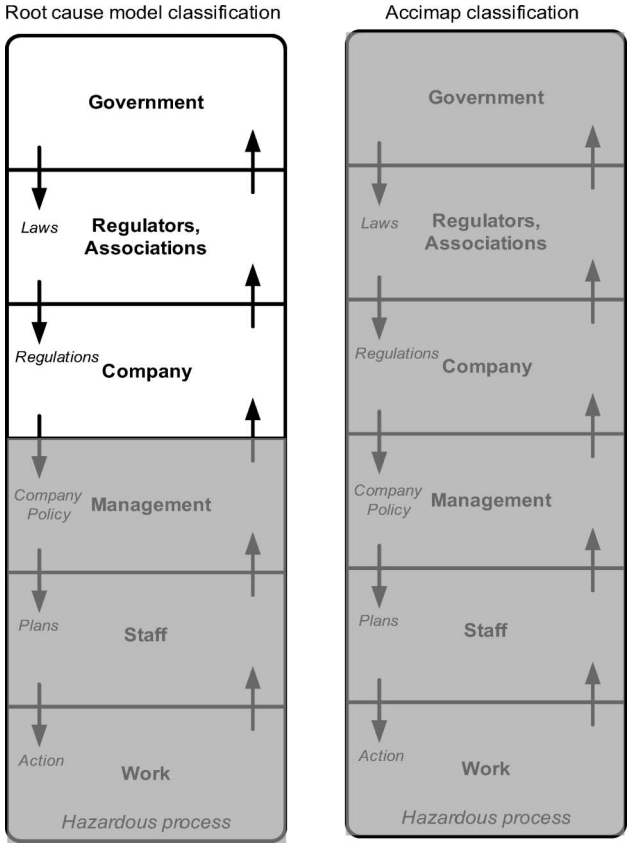

Figure 6. Accimap and root cause model analyses mapped on to Rasmussen's risk management framework.

As discussed earlier, the root cause model has a number of limitations. The root cause taxonomy is narrow and no taxonomies of different failures at the higher managerial system levels are present. Further, no link is made between managerial failures and the root causes identified. The model also pays scant attention to the role of the physical environment and operating conditions in accidents, which is surprising given its development in a led outdoor activity context. Finally, the model lacks a sound theoretical underpinning.

Thus, it is concluded that the Accimap method is the most suited of the two for the analysis of safety-compromising accidents and incidents in the led outdoor activity domain. Of particular relevance is the focus of the Accimap approach on the higher managerial, local authority, regulatory and government failures involved. Failures at these higher systemic levels are now accepted to play a critical role in accidents and incidents in the safety-critical domains (e.g. Reason 1990, Rasmussen 1997, Hollnagel 2004). Applying methods that do not highlight such failures is likely to lead to the development of inappropriate remedial measures and countermeasures. Reason (1997) and Dekker (2002), for example, discuss the problem of proposing front line operator focused countermeasures that are inappropriate and ignore the wider systemic failures involved.

The study presented does have some key limitations. First, only one accident was subject to analysis. This was primarily a function of the lack of detailed data available on accidents and incidents in the led outdoor activity domain, which limited the number of cases that could be analysed to the level of detail required for accident model comparison. The Lyme Bay incident was suitable for detailed analysis due to the quality of the data available in the inquiry report. A preliminary investigation of the data associated with a series of other accidents provided by the project stakeholder indicated that the data would not support such detailed analyses. Second, only two accident models were compared. As cited previously, various models exist, both within the general ergonomics literature (e.g. Reason 1990, Perrow 1999, O'Hare 2000, Leveson 2004) and within the led outdoor activity and adventure tourism literature (e.g. Meyer 1979 cited in Davidson 2004, Brackenreg 1999, Davidson 2007, Bentley and Page 2008). However, the Accimap approach was chosen due to its current popularity in human factors and ergonomics circles (e.g. Cassano-Piche *et al.* 2009, Jenkins *et al.* 2010), whilst the root cause model was chosen as it is the most recent model to be developed specifically for the led outdoor activity domain.

Compared to other domains in which accidents and incidents are a significant problem, only limited theoretically underpinned research focusing on accident causation has been undertaken within the led outdoor activity domain. With participation on the increase, and safety-compromising accidents and incidents continuing to occur, further research is required. The analysis of accidents and incidents via Accimap represents one approach that can enhance safety through increasing activity providers' understanding of the nature of accidents and informing the development of appropriate strategies and countermeasures designed to prevent or mitigate future accidents and incidents. Other pertinent lines of work recommended are the development of standardised accident and incident reporting systems, the development of a led outdoor activity specific accident and incident analysis methodology, the development of a universal database of outdoor activity accidents and incidents and the development of taxonomies of system failure and human error. In particular, the development of failure taxonomies for each of the levels used by the Accimap method may be one line of research that enhances the usability of the approach for analysing accidents and incidents in the led outdoor activity domain.

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